

Module 7: Communication — Making Data Tell Its Story

Learning Objectives

1. Learn to **communicate mathematical findings** clearly through graphs, charts, and written explanations.
2. Understand how data can be **presented honestly or misleadingly** and develop skills to tell the difference.
3. Practice **translating between** numbers, visuals, and words — the three languages of mathematical communication.

Introduction

You've done the detective work. You've collected data, spotted patterns, tested predictions, and updated your models. Now comes a skill that's just as important as all of that: **communicating what you found**.

A brilliant mathematical insight that nobody can understand is useless. A dataset that isn't visualized clearly might as well not exist. The best mathematicians and scientists are not just good at finding answers — they're good at explaining them so other people can understand and use them.

And here's the thing: how you present data can change what people believe. A misleading graph can make people think something is true when it's not. A well-designed chart can make a complex pattern instantly clear. Mathematical communication is power — and with power comes responsibility.

Key Concepts

1. Three Languages of Math

Mathematical ideas can be expressed in three "languages," and a good communicator can translate between them:

Numbers (the raw data): "The basketball team scored 45, 52, 38, 61, and 55 in their last five games."

Visuals (graphs and charts): A bar graph showing the scores, making it instantly clear that game 3 was an outlier and there's generally an upward trend.

Words (explanation): "The team's scores have been improving, with one bad game in the middle. Their average is about 50 points per game."

Each language has strengths. Numbers are precise. Visuals reveal patterns. Words provide context and interpretation. The strongest mathematical communication uses all three together: show the data, visualize the pattern, and explain what it means.

When you "show your work" on a math problem, you're communicating in all three languages: the numbers (your calculations), the visuals (diagrams or graphs you drew), and the words (your reasoning about why you chose your approach).

2. Honest vs. Misleading Data Presentation

The same data can be presented in ways that are honest or misleading. Here are some common tricks to watch for:

Manipulated scales: A graph showing a company's stock price from \$99 to \$101 looks like a dramatic spike. The same data on a \$0 to \$200 scale looks completely flat. The scale changes the story.

Cherry-picked time ranges: A graph showing temperatures rising from January to July suggests warming. But it's just normal seasonal change — if you showed the full year, you'd see temperatures drop again from July to December.

Missing context: "Our product is 200% better!" Better than what? Measured how? Without context, a statistic is just a number wearing a costume.

Misleading chart types: Using a 3D pie chart makes some slices look bigger than they are due to perspective. Using a bar chart with pictures (like stacks of money) can distort proportions.

Being a Math Detective means not just creating honest visualizations — it means being able to spot when other people's visualizations are misleading.

3. Showing Your Reasoning

In math class, "showing your work" often feels like a pointless chore. But there's a real reason for it: your work IS your communication.

When you write out your steps, you're doing two things. First, you're leaving a trail that lets someone else (or future you) follow your reasoning and check it. Second, you're clarifying your own thinking — often you catch your own errors in the act of writing things down.

Good mathematical communication follows a simple structure: 1. **What's the question?** (What are you trying to find out?) 2. **What's your approach?** (What method or model are you using, and why?) 3. **What are the steps?** (Show your work clearly) 4. **What's the answer?** (State it clearly, with units) 5. **Does it make sense?** (A quick reasonableness check)

This structure works for a math test, a science project, or a presentation to your class. It's the universal format for mathematical communication.

Try It!

Data Storyteller. Collect a simple dataset about your class or school (favorite foods, average homework time, number of siblings — anything you can survey). Present the same data three ways: (1) A table of numbers, (2) A chart or graph, (3) A written paragraph explaining what the data shows. Then create a SECOND graph of the same data that tells a different (or misleading) story by changing the scale, leaving out data points, or choosing a different chart type. Present both versions and challenge your classmates to spot which one is honest and which one is manipulated.

Summary

Mathematical communication uses **three languages** — numbers, visuals, and words — and the strongest communication uses all three. Data can be presented **honestly or misleadingly**, and a Math Detective needs to both create honest visualizations and spot dishonest ones.

Showing your reasoning isn't just a school requirement — it's how mathematical ideas get shared, verified, and built upon. Final module ahead: **Planning** — using your mathematical skills to make smarter decisions about the future.